RWANDA POLYTECHNIC

MUSANZE COLLEGE

FARMVISION AI™

Agricultural Intelligence Operating System (AIOS)

Tagline

"Transforming Agricultural Data Into Intelligent Decisions, Automated Actions, and Sustainable Productivity."

Team Member 1

IRAVUGA Chrismas

- RP Musanze College
- Advanced Diploma in Electrical Automation Technology
- Year 3
- Registration No: 24RP02129

Role:

- AI Systems Integration Engineer
- IoT Architect
- Automation Engineer
- Hardware Systems Engineer

Team Member 2

IRADUKUNDA RUDASINGWA Emmanuel Bienvenu

- RP Musanze College
- Advanced Diploma in Information Technology
- Year 2
- Registration No: 25RP16047

Role:

- Full Stack Software Developer
- Data Engineer
- Cloud Platform Engineer

EXECUTIVE SUMMARY

Agriculture remains the backbone of Rwanda's economy, providing livelihoods for millions of citizens while supporting national food security and economic growth.

Despite significant progress, many farmers, cooperatives, and agricultural institutions continue to face challenges including:

- Low productivity
- Resource wastage
- Climate-related risks
- Livestock and crop losses
- Limited access to real-time intelligence
- Delayed decision-making
- Insufficient integration of agricultural data sources

At the same time, vast amounts of agricultural data already exist across:

- Farm operations
- Sensor networks
- Weather systems
- Climate platforms
- Agricultural surveys
- Market information systems
- Government statistical databases

However, these datasets often operate independently, limiting their potential value.

FarmVision AI™ addresses this challenge by introducing a new category of agricultural technology:

Agricultural Intelligence Operating System (AIOS)

Unlike conventional agricultural systems that only monitor or report information, FarmVision AI continuously:

OBSERVE
↓
UNDERSTAND
↓
PREDICT
↓
RECOMMEND
↓
AUTOMATE
↓
LEARN

By combining Artificial Intelligence, Big Data, IoT, Computer Vision, Predictive Analytics, Climate Intelligence, and Smart Automation, FarmVision AI transforms agricultural data into actionable intelligence capable of improving productivity, sustainability, and food security.

PROJECT VISION

Vision

To become Africa's leading Agricultural Intelligence Operating System, empowering every farmer, cooperative, agribusiness, researcher, and policymaker with intelligent decision-support tools powered by Artificial Intelligence and real-time agricultural insights.

Mission

To transform agricultural data into actionable intelligence that increases productivity, strengthens climate resilience, improves sustainability, and enhances food security.

PROBLEM STATEMENT

The Agricultural Decision Problem

Every day, agricultural stakeholders make critical decisions:

- When to plant
- When to irrigate
- When to harvest
- How much feed to provide
- How much water is required
- Which environmental risks exist
- How to maximize productivity

Most of these decisions are still based on:

- Experience
- Assumptions
- Delayed information
- Manual observations

The consequences include:

- Reduced productivity
- Increased operational costs
- Wasted resources
- Crop losses
- Livestock losses
- Reduced food security

National Challenge

Rwanda requires modern agricultural solutions capable of:

- Increasing productivity
- Improving climate resilience
- Enhancing resource efficiency
- Supporting evidence-based policies
- Strengthening food security

Current solutions typically address only one problem area:

- Weather applications
- Irrigation systems
- Livestock monitoring systems

- Farm management software

FarmVision AI integrates these capabilities into one intelligent ecosystem.

THE FARMVISION AI ECOSYSTEM

FarmVision AI is not:

❌ A dashboard

❌ A farm management application

❌ A smart feeder

❌ An IoT monitoring project

FarmVision AI is:

✅ An Agricultural Intelligence Operating System

The platform brings together:

Farmers
+
Cooperatives
+
Agronomists
+
Researchers
+
Government Institutions
+
IoT Infrastructure
+
Artificial Intelligence
+
Climate Intelligence
+
Automation

into a unified digital agricultural ecosystem.

PLATFORM ECOSYSTEM

1. Farmer Intelligence Platform™

Provides:

- Farm registration
- Crop management
- Livestock management
- AI assistant
- Productivity scoring
- Environmental monitoring
- Production analytics
- Automated alerts

2. Cooperative Intelligence Platform™

Provides:

- Member management
- Production planning
- Resource optimization
- Regional analytics
- Disease surveillance
- Cooperative performance monitoring

3. Government Intelligence Platform™

Provides:

- Food security monitoring
- National production analytics
- Climate risk analysis

- Agricultural forecasting
- District performance comparisons
- Policy decision support

4. Research & Innovation Platform™

Supports:

- Universities
- Innovation centers
- Researchers
- Agricultural institutions

through advanced analytical tools and agricultural intelligence services.

5. Agricultural Marketplace™

Connects:

- Farmers
- Buyers
- Suppliers
- Cooperatives
- Agribusinesses

for more efficient agricultural commerce.

HARDWARE ECOSYSTEM

One of the key differentiators of FarmVision AI is its integration of intelligent hardware devices with cloud-based agricultural intelligence.

FarmVision Feeder™

Status:

Operational Working Prototype

Purpose

A smart livestock automation unit designed to automate feeding, watering, monitoring, and environmental control.

Functions

- Automated feeding
- Automated watering
- Feed monitoring
- Water monitoring
- Environmental monitoring
- SMS notifications
- Livestock automation

Hardware Components

- ESP32
- 2 × MG90S Servo Motors
- HX711 Load Cell System
- DHT22 Sensor
- DS3231 RTC Module
- SIM800L GSM Module
- LCD Interface
- Relay Control System
- Water Pump
- Fan Control
- Lighting Control
- Ultrasonic Sensors

Demonstrated Capabilities

Automatic Feeding
Automatic Watering
Temperature Monitoring

Humidity Monitoring
SMS Alerts
Actuator Control
Data Collection
Smart Automation

FarmVision Node™

Status:

Architecture Complete | Future Hardware Expansion

Purpose

Smart farm monitoring node for data acquisition.

Future Sensor Integration

- Soil moisture sensors
- Water quality sensors
- Environmental sensors
- Farm telemetry devices

Functions

- Environmental monitoring
- Data collection
- IoT communication
- Agricultural intelligence generation

FarmVision Climate™

Status:

Digital Prototype

Functions

- Climate tracking
- Environmental monitoring
- Weather intelligence
- Risk prediction

Intelligence Outputs

- Drought risk prediction
- Heat stress monitoring
- Climate impact analysis
- Environmental recommendations

FarmVision Vision™

Status:

Prototype Architecture

Technology

- ESP32-CAM
- Computer Vision Services
- AI-powered Image Analytics

Functions

- Livestock observation
- Crop monitoring
- Activity tracking
- Security monitoring
- Risk detection

INTELLIGENCE ENGINE

FarmVision Intelligence Engine™

The core AI engine transforms raw data into decisions.

Yield Prediction Engine

Predicts:

- Future yields
- Production risks
- Expected harvest performance

Inputs:

- Historical production
- Climate conditions
- Farm records
- Environmental data

Disease Prediction Engine

Identifies risks before outbreaks occur.

Outputs:

- Disease probability
- Risk scoring
- Prevention recommendations

Feed Intelligence Engine

Predicts:

- Feed demand
- Future shortages
- Consumption trends
- Resource planning requirements

Water Intelligence Engine

Analyzes:

- Water demand
- Consumption behavior
- Water shortages
- Irrigation schedules

Climate Intelligence Engine

Analyzes:

- Environmental conditions
- Climate threats
- Weather impacts
- Climate adaptation actions

Farm Scoring Engine

Generates intelligent farm performance scores based on:

- Productivity
- Resource efficiency
- Sustainability
- Operational performance

DIGITAL TWIN TECHNOLOGY

FarmVision AI creates a digital representation of every farm.

The Digital Twin simulates:

- Feed consumption
- Water consumption
- Weather changes
- Environmental risks
- Production performance
- Resource shortages

This enables predictive decision-making before problems occur in the real environment.

DATA ECOSYSTEM

FarmVision AI integrates five major categories of data.

Agricultural Statistics

Including:

- Agricultural surveys
- Livestock statistics
- Food security indicators
- Production datasets

Climate Data

Including:

- Rainfall
- Temperature
- Humidity
- Seasonal forecasts

IoT Data

Collected from:

- Sensors
- Smart devices
- Monitoring systems
- Intelligent hardware

Market Intelligence

Including:

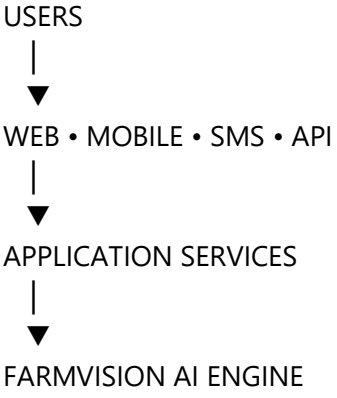
- Commodity prices
- Supply patterns
- Demand trends
- Marketplace analytics

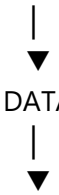
Farm Data

Generated through:

- Farmers
- Agronomists
- Cooperatives
- Field operations

TECHNICAL ARCHITECTURE





Application Layer

Includes:

- Farm Management
- Livestock Management
- Crop Intelligence
- Marketplace
- Government Analytics
- Cooperative Services
- Reporting Engine

Technology Stack

Frontend

- React
- Next.js
- TypeScript
- Flutter

Backend

- Node.js
- Express.js
- TypeScript

Database

- PostgreSQL
- Time-Series Storage

IoT

- MQTT
- REST APIs
- ESP32 Platform

Artificial Intelligence

- Machine Learning
- Predictive Analytics
- Recommendation Systems
- Decision Intelligence Engines

BUSINESS MODEL

Revenue Stream 1

Software Subscription

Target customers:

- Farmers
- Cooperatives
- Agribusinesses

Revenue Stream 2

Government Analytics Services

- Forecasting reports
- Policy intelligence
- Decision-support services

Revenue Stream 3

Marketplace Commission

Revenue generated from platform transactions.

Revenue Stream 4

Hardware Products

Including:

- FarmVision Feeder™
- FarmVision Node™
- FarmVision Climate™
- FarmVision Vision™

Revenue Stream 5

Agricultural Intelligence Services

Advanced:

- Forecasting
- Analytics
- Research intelligence
- Data services

EXPECTED IMPACT

Farmers

- Higher productivity
- Reduced losses

- Lower costs
- Improved decision-making

Cooperatives

- Better planning
- Improved coordination
- Resource optimization

Government

- Food security intelligence
- Agricultural forecasting
- Evidence-based policymaking

Rwanda

- Increased productivity
- Improved food security
- Rural economic development
- Agricultural digital transformation

COMPETITIVE ADVANTAGE

Traditional Solutions

Collect Data
↓
Display Data

FarmVision AI

Collect Data
↓
Analyze Data
↓
Predict Future Events
↓
Recommend Actions
↓
Automate Responses
↓
Continuously Learn

Most agricultural solutions solve a single problem.

FarmVision AI solves the entire agricultural decision cycle.

CONCLUSION

FarmVision AI™ is not a farm monitoring application.

FarmVision AI™ is not an IoT dashboard.

FarmVision AI™ is not a smart feeder.

FarmVision AI™ is an Agricultural Intelligence Operating System (AIOS).

By integrating Artificial Intelligence, Big Data, IoT, Computer Vision, Climate Intelligence, Smart Automation, Digital Twin Technology, and Agricultural Analytics into a unified ecosystem, FarmVision AI creates a scalable platform capable of transforming agricultural decision-making in Rwanda and across Africa.

The current working prototype, **FarmVision Feeder™**, demonstrates the platform's core automation capabilities through intelligent feeding, watering, environmental monitoring, and real-time farm control. Combined with the broader FarmVision AI ecosystem, this forms the foundation for a next-generation agricultural operating system designed to improve productivity, strengthen food security, and drive sustainable agricultural transformation.

Submitted By

IRAVUGA Chrismas
RP Musanze College
Advanced Diploma in Electrical Automation Technology

IRADUKUNDA RUDASINGWA Emmanuel Bienvenu
RP Musanze College
Advanced Diploma in Information Technology

Competition Track: Agricultural Productivity

Category: Artificial Intelligence • Big Data • IoT • Computer Vision • Agricultural Intelligence • Smart Automation

TEAM STRUCTURE

1. IRAVUGA Chrismas

Co-Founder & Chief Systems Integration Engineer

Primary Responsibility: Make the Farm Work

You own everything that exists physically and everything that connects the physical world to the digital world.

A. COMPLETE HARDWARE WIRING

Power Section

Design:

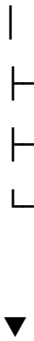
Plain Text

220VAC

|

▼

12V SMPS



ESP32 Connections

I2C Devices

Plain Text

GPIO21 → SDA

GPIO22 → SCL

Connected to:

Plain Text

LCD 20x4

RTC DS3231

Shared Bus.

HX711

Plain Text

HX711 DT → GPIO32

HX711 SCK → GPIO33

Purpose:

Plain Text

Feed Weight Measurement

Feed Level Sensor

Plain Text

HC-SR04 #1

TRIG → GPIO4

ECHO → GPIO35

Purpose:

Plain Text

Feed Tank Monitoring

Water Level Sensor

Plain Text

HC-SR04 #2

TRIG → GPIO5

ECHO → GPIO26

Purpose:

Plain Text

Water Tank Monitoring

DHT22

Plain Text

DATA → GPIO14

Purpose:

Plain Text

Temperature

Humidity

Servo Motors

Feed Gate

Plain Text

MG90S #1

Signal → GPIO19

Feed Dispenser

Plain Text

MG90S #2

Signal → GPIO18

Relay Outputs

Plain Text

GPIO13 → Fan Relay

GPIO27 → Lamp Relay

GPIO23 → Pump Relay

Buzzer

Plain Text

GPIO25

GSM

Plain Text

GPIO16 → GSM TX

GPIO17 → GSM RX

Purpose:

Plain Text

SMS Alerts

Main Areas

A. Hardware Architecture

Design:

FarmVision Feeder™

FarmVision Node™

FarmVision Climate™

FarmVision Vision™

Deliverables:

Electrical Schematics

Hardware Selection

Power Architecture

Sensor Architecture

Communication Architecture

B. Embedded Systems

Build:

ESP32 Firmware

Sensor Drivers

Servo Control

Relay Control

RTC Control

GSM Communication

Device Logic

C. Automation Engineering

Develop:

Automatic Feeding

Automatic Watering

Climate Monitoring

Alarm Management

Actuator Control

Safety Logic

D. IoT Integration

Implement:

MQTT Communication

Device Registration

Data Transmission

Cloud Connection

E. Prototype Construction

Responsible for:

Mechanical Assembly

Wiring

Testing

Calibration

Troubleshooting

YOUR SUCCESS TESTS

The judges must physically see:

Test 1: Automatic Feeding

Input:

12:00 PM Feeding Time

Expected:

Servo Opens

Feed Dispensed

Weight Recorded

Servo Closes

Data Saved

Pass = Working

Test 2: Feed Low Alert

Input:

Feed Tank <20%

Expected:

LCD Alert

Dashboard Alert

SMS Alert

Pass = Working

Test 3: Water Low

Input:

Low Water Level

Expected:

Pump Starts

Level Restored

Pump Stops

Pass = Working

Test 4: High Temperature

Input:

Temperature > 30°C

Expected:

Fan ON

Alert Generated

Pass = Working

Test 5: Sensor Monitoring

Expected Display:

Temperature

Humidity

Feed Level

Water Level

Weight

Real-time updates.

Pass = Working

2. IRADUKUNDA RUDASINGWA Emmanuel Bienvenu

Co-Founder & Platform Engineering Lead

Primary Responsibility: Make the Intelligence Work

Own everything from dashboard to AI.

A. Farmer Platform

Develop:

Farmer Dashboard

Farm Registration

Animal Registration

Crop Management

Reports

B. Backend Development

Develop:

Authentication

API Services

Database

Application Logic

C. Data Engineering

Implement:

Data Collection

Data Processing

Data Analytics

Data Warehousing

D. AI Platform

Build:

Feed Intelligence

Water Intelligence

Yield Prediction

Disease Prediction

Climate Intelligence

E. National Intelligence Dashboard

Build:

Government Dashboard

Cooperative Dashboard

Research Dashboard

F. Marketplace

Build:

Buyers

Suppliers

Farmer Connections

SOFTWARE SUCCESS TESTS

Judges must see:

Test 1: Farm Registration

Input:

Create Farm

Expected:

Farm Saved

Farmer Linked

Pass = Working

Test 2: Device Data Reception

Input:

ESP32 Sends Data

Expected:

Dashboard Updates

Database Updates

Pass = Working

Test 3: AI Recommendation

Input:

Temperature = 32°C

Humidity = 85%

Expected:

Heat Stress Risk

Recommended Actions

Pass = Working

Test 4: Yield Prediction

Input:

Farm Data

Climate Data

Historical Records

Expected:

Predicted Yield

Risk Score

Pass = Working

Test 5: Digital Twin

Input:

Feed Reduced

Water Reduced

Expected:

Virtual Farm Changes

Risk Prediction Generated

Pass = Working

WHAT I RECOMMEND YOU FOCUS ON NOW

Because you have limited funds:

Chrismas (You)

Focus 80% on:

FarmVision Feeder™

Automatic Feeding

Automatic Watering

Environment Monitoring

SMS Alerting

ESP32 Integration

Make it PERFECT.

A working prototype impresses judges more than ten unfinished devices.

Emmanuel

Focus 80% on:

Dashboard

Backend

AI Recommendation Engine

Digital Twin

Government Dashboard Mockup

DEMO FLOW FOR JUDGES

Step 1

Open Dashboard

Show:

Farm Registered

Step 2

Show Real Hardware

ESP32

Sensors

Servos

LCD

Step 3

Press DEMO MODE

System shows:

Feeding Time

Servo rotates.

Step 4

Simulate Feed Low

Dashboard:

Feed Level = 10%

Alert appears.

Step 5

Simulate High Temperature

Temperature = 35°C

Fan activates.

Step 6

Show AI

Feed Consumption Trend

Predicted Feed Shortage

Recommended Refill Date

FINAL MESSAGE TO JUDGES

Chrismas demonstrates the physical intelligence layer.

Observe

Measure

Control

Automate

Emmanuel demonstrates the digital intelligence layer.

Analyze

Predict

Recommend

Optimize

Together, both layers become:

FARMVISION AI™

Agricultural Intelligence Operating System

Observe

↓

Understand

↓

Predict

↓

Recommend

↓

Automate

↓

Learn

That is the strongest story and division of responsibilities for your proposal and presentation.

TEAM STRUCTURE

IRAVUGA Chrismas

Co-Founder

AI Systems Integration Engineer

IoT Architect

Automation Engineer

Hardware Systems Engineer

Responsibilities

- Hardware architecture
- Embedded systems development
- IoT integration
- Automation design
- ESP32 firmware development
- Sensor integration
- Relay and actuator control
- GSM communication
- System integration
- Prototype deployment

IRADUKUNDA RUDASINGWA Emmanuel Bienvenu

Co-Founder

Full Stack Software Developer

Data Engineer

Cloud Platform Engineer

Responsibilities

- Backend systems development
- Frontend development
- Mobile application development
- Cloud architecture
- API development
- Database systems
- Data engineering
- Analytics platforms
- Dashboard development
- Platform integration